

RAD Chapter 7 - Quality Assurance and Implementation

Simple English Short Notes

This note summarizes Chapter 7 topics: Software Quality, SQA, Data Quality, TQM, Six Sigma, Structured Walkthrough, and Software Evaluation Techniques.

1. Software Quality

Software quality means how well software satisfies users and project requirements. Good quality software should work correctly, perform well, be easy to learn, easy to use, and easy to maintain. It should also follow development standards.

Main points: customer satisfaction, correct functions, good performance, easy maintenance, and standard development practices.

2. Software Quality Assurance (SQA)

SQA is a planned process used to make sure software engineering activities follow defined standards. It checks whether processes, methods, activities, and work products are properly monitored and controlled.

Simple idea: SQA tries to prevent quality problems before the final product is delivered.

Term	Meaning
Software Quality	How good the software is for users and requirements.
SQA	A systematic process that checks whether software work follows standards.

3. Data Quality

Data quality means data should be accurate, consistent, complete, reliable, and useful. If data is poor, the system output will also be poor.

Key strategies for ensuring data quality:

Data validation: Use rules to check accuracy and completeness.

Data cleansing: Correct wrong data and standardize data formats.

Data monitoring: Continuously check data and fix issues early.

Documentation: Maintain data dictionaries and metadata clearly.

User training: Teach users how to enter and manage good quality data.

4. Total Quality Management (TQM)

TQM is an organization-wide approach for continuous improvement. It focuses on customer satisfaction, teamwork, leadership, employee involvement, and improving quality at every stage.

Important idea: In TQM, quality is not a one-time activity. It is a continuous journey toward perfection.

5. Six Sigma

Six Sigma is a quality improvement approach that aims to reduce defects and variation in a process. It was originally developed by Motorola in the 1980s. The word sigma refers to standard deviation, which measures variation.

Goal: reduce defects as much as possible and make the process close to perfect.

Six Sigma is used for products, services, and processes. It is a top-down approach, meaning it is strongly supported by top management.

Six Sigma Roles

Role	Description
Black Belt	Project leader who leads Six Sigma projects.
Green Belt	Project team member who supports Six Sigma activities.
Master Black Belt	Highly experienced expert who supports many project teams.

Steps of Six Sigma

1. Define the problem
2. Observe the problem
3. Analyze the causes
4. Act on the causes
5. Study the results
6. Standardize the changes
7. Draw conclusions

6. Structured Walkthrough

A structured walkthrough is a **review meeting** where **team members check a completed part of a system**, program, or subsystem. The aim is to find problems early and improve quality.

Walkthroughs use peer reviewers. They point out issues, and the programmer or analyst can make suitable changes.

People involved in a structured walkthrough:

- Person responsible for the part being **reviewed**
- Walkthrough **coordinator**
- **Programmer** or analyst peer
- **Note taker** who records suggestions

When can it be done? When a portion of coding, system, or subsystem is complete.

7. Evaluation Techniques

Software evaluation means checking software to decide whether it is suitable, effective, and high quality. Evaluation happens throughout the SDLC. Analysts, management, and users give feedback to improve the system.

Common software evaluation techniques:

Surveys and questionnaires: Collect feedback from users and stakeholders about satisfaction, usability, and expectations.

Functionality testing: Check whether the software performs required functions correctly.

Non-functional testing: Check performance, security, usability, reliability, and similar quality factors.

Cost-benefit analysis: Compare software cost with the value and benefits it gives.

Peer reviews and inspections: Experts review code, design, and documents to find issues.

Benchmarking: Compare software with industry standards or similar products.

Case studies and success stories: Study real examples of organizations that used the software.

8. Exam Quick Answers

Q1: State three key strategies to ensure data quality.

Data validation, data cleansing, and data monitoring. Other answers: documentation and user training.

Q2: List the people involved in structured walkthroughs.

The person responsible for the reviewed part, walkthrough coordinator, programmer or analyst peer, and note taker.

Q3: What is SQA?

SQA is a systematic process that ensures software engineering processes and work products follow defined standards.

Q4: What is the goal of Six Sigma?

The goal of Six Sigma is to reduce variation and eliminate defects in a process.

9. One-Page Revision Summary

Topic	Very short meaning
Software Quality	Software satisfies users, requirements, performance, and standards.
SQA	Process to ensure software work follows standards.
Data Quality	Data is accurate, consistent, reliable, and useful.
TQM	Continuous improvement with customer focus and teamwork.
Six Sigma	Quality method to reduce defects and variation.
Structured Walkthrough	Peer review meeting to find problems early.
Evaluation Techniques	Methods used to check software suitability and quality.